

Continuous Integration (CI) Pipeline

Project: software-architecture-labs | **Feature:** Group / Ungroup figures (JHotDraw 7)

What CI is

Continuous Integration is the practice of merging every developer's work into a shared mainline frequently, where each merge automatically triggers a build and test run. This catches integration errors early instead of at release time.

The pipeline

The pipeline is defined by a single GitHub Actions workflow: `.github/workflows/ci.yml`. It runs automatically on:

- every **push** to the `main` branch, and
- every **pull request** targeting `main`.

What it does

Step	Action	Purpose
1	<code>actions/checkout@v4</code>	Check out the repository on the runner.
2	<code>actions/setup-java@v4</code> (Temurin JDK 8 , Maven cache)	Provide a JDK that accepts JHotDraw 7's Java 1.6 source level, and cache <code>~/.m2</code> for speed.
3	<code>mvn -B package --file jhotdraw/jhotdraw7/pom.xml</code>	Build the JHotDraw 7 Maven module. The <code>package</code> goal runs <code>compile → test → package</code> , so it both builds and runs the test phase in one step.

Notes on tests

The JHotDraw 7 module (`org.jhotdraw:jhotdraw:7.7.0`) ships **no unit tests** (`src/test` does not exist) and its `pom.xml` skips the Surefire plugin. The test phase therefore runs but executes 0 tests. A real change to the Group/Ungroup feature would add e.g. a `GroupFigureTest` under `src/test/java`, and Surefire would then execute it automatically through the same `package` step — no workflow change needed.

GitHub Packages / `.maven-settings.xml`

The lab mentions creating a `.maven-settings.xml` to pull shared jars from GitHub Packages. This project does **not** need one: the `jhotdraw7/pom.xml` has no external dependencies, so there are no private packages to authenticate against. If a dependency on a GitHub Packages artifact were added, the workflow would pass `--`

settings .maven-settings.xml to Maven and the file would hold the registry URL plus a `${GITHUB_TOKEN}` server entry.

Result

Once pushed, the workflow appears under the repository's **Actions** tab and runs on every push/PR, giving a green/red build status that gates integration.